

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Built production LLM agent pipelines across **3 providers and 8+ models** (Claude, GPT-4o, Gemini) at an AI startup -- 394 commits, 7 deployments. RAG with per-tenant data isolation (Glance AI), event-driven **agentic orchestration at 2,000 async req/sec** (AlgoPulse + Temporal + Kafka), and a vendor-neutral LLM eval harness tracking hallucination rate, p50/p95 latency, and cost/call across 2,000 examples. Stack: **Python, PyTorch, LangChain, Hugging Face, vector databases, AWS/GCP**. Ready to build AI agents for Pebble's vehicle intelligence platform.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Built **multi-model AI agent tooling across 3 providers and 8+ models** (Anthropic Claude, OpenAI GPT-4o, Google Gemini) via MCP; designed LLM pipelines with structured outputs, prompt engineering, and multi-model routing across 7 production deployments. **394 commits**.
- Shipped 10+ backend features using **TypeScript, Node.js, Express, Prisma ORM, PostgreSQL, Redis, Temporal.io** workflows, Pusher real-time event pipelines, Auth0, Docker, Sentry, and Playwright; implemented document processing pipelines (PDF gen, email parsing via Microsoft Graph API, S3 storage).

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built AI-driven claim automation workflow (OCR, NLP, ML) improving accuracy 25%; automated deployments on **AWS and GCP** with Grafana + Sentry observability and red-teaming support.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka** + Spark + Hadoop pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

LLM Eval Harness 2025

Vendor-neutral benchmark comparing Claude, GPT-4o, and Gemini on 2,000-example frozen test set. Tracked **ExactMatch, grounded hallucination rate, p50/p95 latency, and cost/call** with bootstrap significance testing and McNemar test. Streamlit dashboard for per-model failure analysis.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

Glance AI -- Privacy-First RAG System 2025

RAG system for fintech/healthcare environments with **per-tenant data isolation at the vector retrieval layer**, preventing context leakage and LLM hallucinations. Deterministic output architecture for compliance-ready regulated data environments.

Python, LLMs, Qdrant, Vector Databases, RAG

VectorLift -- Semantic Search & Ranking Engine 2026

Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant vector store; dense bi-encoder retrieval (sentence-transformers fine-tuned on MS MARCO) + cross-encoder re-ranker; evaluated 6 pipeline configs with NDCG@10, MRR@10, MAP, and bootstrap significance testing.

Python, PyTorch, Sentence-Transformers, Elasticsearch, Qdrant, Kafka, FastAPI, Streamlit, Docker

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, Big Data Infra, Data Warehousing, Algorithms

SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, C

AI/ML: OpenAI API, Claude API, Gemini API, LangChain, Hugging Face, PyTorch, TensorFlow, RAG, Qdrant, Elasticsearch, MCP, LLM Evals

Stack: Node.js, React, FastAPI, Temporal.io, Kafka, Spark, Prisma ORM, Express

Infra: Docker, AWS, GCP, PostgreSQL, Redis, MongoDB, Prometheus, Grafana, GitHub Actions, Sentry

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)